

Taichi Uchida.

Fourth-year aerospace engineering student at Kyoto University, researching mission design, trajectory optimization, and robotics.

uchida.taichi.84f@st.kyoto-u.ac.jp

Education

2027.04	SOKENDAI · Planned enrollment Five-year doctoral program, Space and Astronautical Science. Graduate Institute for Advanced Studies. Naoya Ozaki's Mission Design Lab.
2023.04 — 2027.03	Kyoto University Fourth-year undergraduate in Engineering Science (Aerospace Engineering). Graduation expected March 2027.
2020 — 2023	Seiko Gakuin High School Kanagawa, Japan
2018 — 2020	Pathways School Gurgaon IB Middle Years Programme · Haryana, India
2017 — 2018	Seiko Gakuin Junior High School Kanagawa, Japan

Research

2025.08 —	JAXA / ISAS · Outer-planet mission design and trajectory optimization Multi-point observations with small outer-planet probes, and low-thrust trajectory optimization with Julia and DDP.
2026.04 —	KYOTO UNIVERSITY · Model predictive control with a Transformer A Transformer-based dynamics model integrated with nonlinear MPC for real-time autonomous tracking on an RC car.
2025.10 —	KYOTO UNIVERSITY · Reinforcement learning and control for lunar robots Reinforcement learning, inverse kinematics, and sim-to-real environments for modular multi-legged lunar robots.
2025.02 — 2025.04	DFKI · BREMEN · 7-DOF robot arm control Simulation environments and impedance / position controllers for a 7-DOF arm, with space-debris capture as the research context.
2025 —	KYOTO UNIVERSITY · TRACK & FIELD · Athletics record management web app A custom React and Firebase application for the Kyoto University Medical School Track & Field Team.
2025.03 —	TLEEZ INC. · Russian grammatical error correction AI-based grammatical error correction and training data generation for Russian language learning.
2024.08 —	GEORGIA TECH · Metagenomic analysis of microbial communities Metagenomic analysis of microbial communities involved in heavy metal contaminant stabilization.
2020.12 — 2023.03	TOKYO TECH · ELSI · Isolation and identification of halophilic archaea Isolation, identification, and phylogenetic analysis of halophilic archaea in rock salt from around the world.

2023.09 — 2024.08

Microbial metabolism · Kyoto University

Wet-lab research on hydrogen and oxygen utilization at the Atomi Lab.

2021.08 — 2022.02

Meteorite analysis · Seiko Gakuin

SEM-based composition analysis with UK schools and university researchers.

2020.08 — 2021.08

Artificial gravity · Seiko Gakuin

Prototyped a mechanism and built Arduino-based force sensors for evaluation.

2021

Intelligent water rocket

A rocket with computer-controlled functions and automated parachute deployment.

Presentations & papers

2026.08

Paper submitted

HDDP-based trajectory optimization for low-thrust GTO-to-GEO transfers

70th UKAREN

2026.03

Oral presentation

Mission design for multi-point simultaneous observations by ultra-small outer-planet probes on large overseas missions ↗

SPS 2026

2023.11

Oral & poster

Space medicine education in Japan

69th JSAEM

2022.05

Poster · Award

Isolation and identification of extreme halophiles from global rock salts

JpGU 2022

2022.06

Online presentation

Extreme halophiles from global rock salts

NASA × AGI × High School Students

2022.03

Online presentation

Research on extreme halophiles

Japanese Society of Scientific Fisheries

2022.01

Online presentation

Meteorite composition analysis with electron microscopy

International Research for School

Awards & scholarships

2025

Startup Weekend Space Osaka · 2nd place

Team leader; business plan and prototype in 54 hours.

2024–25	Nakatani Foundation Scholarship Fully funded research at Georgia Tech and DFKI.
2022	JpGU Award for Academic Excellence Research on halophilic archaea.
2022	Japanese Society of Scientific Fisheries Award Extreme halophiles research.
2022	SDGs Tankyu Award · Nihon Ryoko Prize Founded and organized Seiko SDGs Food Ambassador.
2021	Intelligent Water Rocket · Missions Award Completed all competition missions.
2021	SSH Research Symposium Award Artificial gravity research.
2021	Yurikamome Relay Marathon · Overall winner Team competition.
2019	ISSO National Games · Three golds 1st in 800m, 1500m, and 3000m in India.

Activities

2026 . 03	JENESYS2025 · Japan–Korea exchange Selected delegate for an international cultural exchange program.
2025 . 05	Interpreting at the Kyoto International Music Students Festival English–Japanese interpreter and liaison at the Kyoto International Music Students Festival.
2023 —	Mathematics instructor at Tetsuryokukai Advanced mathematics instructor at Tetsuryokukai.
2021	Japan Students Space Forum Founded and organized a forum with more than 100 participants.
2021	Seiko SDGs Food Ambassador Founded a project supporting regional producers through a school festival.
School years	The “Yokohama Banana” project Founded and managed a venture that sold 3,800 chocolate bananas over two days.
2025 / 2023	Space medicine & Moon Village Staff at the 71st JSAEM and 7th Global Moon Village Workshop; organized a university space medicine study tour.
2012 —	Rice farming & Shorinji Kempo Long-term participation in organic rice farming; competed at the national junior-high Shorinji Kempo championship.

Science & international exchange

Founded the Science Club Summit; organized India–Japan student discussions and a Japanese calligraphy event.

2021

Legislative internship

ICAS program at the Tokyo Metropolitan Assembly.

Skills

Programming	Python · Julia · C++ · MATLAB · R · Go
Control & robotics	DDP · NMPC · Drake · Pinocchio · MuJoCo · Isaac Lab · CasADi · ROS2
Machine learning & web development	PyTorch · React · Next.js · Firebase · Docker · Linux · Git
Life science	PCR · DNA extraction · PhyML · Metagenomics
Languages	Japanese (native) · English (TOEFL iBT 100 / EIKEN Grade 1) · Russian (learning)
Qualifications	Model Rocket License Grade 4 · NAUI Open Water Diver · Driver's license (AT)
